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Advancing Surfactant-Based Chemical EOR in High-Salinity, High-Temperature Carbonate Reservoirs of Sultanate of Oman

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Abstract

Carbonate reservoirs, which account for over 60% of global oil reserves, are predominantly oil-wet which limits their oil recovery factors from using conventional primary and secondary methods. In Oman, these reservoirs are further challenged by high reservoir temperatures up to 85°C, brine salinities up to 180,000 ppm, and geological heterogeneity. This study aims to identify and validate wettability-altering surfactants to enhance oil recovery in two such carbonate reservoirs (Natih and Lower Shuaiba) under high-salinity, high-temperature (HSHT) conditions. A two-phase screening methodology was employed. First, a broad analytical pre-screening combined SPE-established criteria with machine learning techniques applied to public-domain performance data to shortlist ten potential surfactants suitable for HSHT environments. These candidates were subjected to detailed experimental evaluation, including compatibility and stability testing, phase behavior analysis under reservoir conditions, and contact angle testing to assess wettability alteration. Reservoir-specific criteria were used to further narrow the list for core flood testing. From the ten initial candidates, four surfactants were identified as promising—two for each reservoir. These surfactants exhibited strong thermal and salinity stability, favorable phase behavior, and showed wettability alteration in contact angle test. Core flood experiments under reservoir conditions were conducted to quantify incremental oil recovery and confirm the effectiveness of the selected surfactants. Good incremental recovery gains were observed compared to baseline waterflooding, validating the surfactants' performance, and supporting their potential for field deployment. The dual-track evaluation—common and reservoir-specific—ensured optimal alignment with the unique properties of each reservoir. This work demonstrates a replicable, cost-effective framework for CEOR surfactant selection in HSHT carbonate reservoirs. The integration of data-driven pre-screening with robust lab testing enables rapid identification of effective, wettability-alteration surfactants tailored to specific geological and operational conditions. The approach not only enhances oil recovery but also considers environmental and economic sustainability. The methodology provides a valuable reference for similar EOR applications across the Middle East and other carbonate-reservoir dominated oil regions globally.

Introduction

Globally, > 60 % of original oil in place (OOIP) resides in oil-wet carbonate reservoirs, yet recovery efficiencies rarely exceed one-third under primary depletion and conventional water flooding. Wettability alteration toward water-wet conditions can enhance spontaneous imbibition, lower residual oil saturation, and reduce capillary trapping (Hirasaki, et.al., 2004, Gupta, et.al, 2009, Mohanty, et.al., 2011, Sheng, 2013 and Sharma, et.al, 2013). Non-ionic and amphoteric surfactants with strong temperature and brine tolerance are attractive for Arabian carbonate EOR, where formation waters commonly exceed 120 k ppm TDS and reservoir temperatures reach 60-85 °C. Petroleum Development Oman (PDO) has already deployed secondary methods in carbonate reservoirs—water flooding, but declining production calls for investigation into enhanced oil recovery methods involving surfactants.

First, large-scale surfactant injection trial for enhanced oil recovery was conducted by BP, across various North Sea fields (Magnus, Fortis, Piper, Beryl, Claymore), in the 1980's (Davison, et.al., 1979, Dymond, et.al., 1988). Marathon Oil company initiated a dilute- surfactant-well stimulation pilot and tested cyclic wettability-altering injection using surfactant in the early 1990s (D. Craig McClure 1989). Surfactant slugs were injected into the oil/water transition zone. Once the surfactant slug was injected, the well was shut in (soak time) for a brief period of time. When the production resumed, the well showed an increase in the recovery of oil. The surfactant has also been used in the Yates field in a number of pilots involving 0.3-0.4% nonionic ethoxy alcohol, anionic ethoxy sulfate and more recently using cationic surfactants (Valdez, et.al., 2020). The surfactant solution injected were prepared with produced water in the concentration of 3100-3880 ppm. The reported results were positive (Yang and Wadleigh, 2000).

Surfactant-based wettability alteration has also been tested for the Middle Eastern carbonate reservoirs. The Mauddud carbonate reservoir in Bahrain has been tested for a combination of Alkaline and Surfactant injection for wettability alteration (Zubari and Sivakumar, 2003). Detailed study on the application of wettability-altering surfactant flooding has been done for Kuwaiti carbonate reservoirs (Jonathan Kuhne, 2023). In Oman, surfactant injection has been carried out for the Kharaib carbonate using diluted surfactant injection showing encouraging results (Al Sinam, et.al, 2016). Further trial expansions were carried out in 2018 and 2024, both reporting the incremental oil recovery (Wang, et.al., 2024). Surfactants have also been extensively studied and tested in Qatar for the Kharaib carbonate reservoir with lab study (Bennetzen et.al, 2014, Pal, et.al, 2018, Pal, et.al, 2019) and field trails (Kumar, et.al. 2024, Kumar, et.al. 2025). The laboratory tests have shown significant incremental oil gain due to the surfactant injection, which was confirmed with two field trials to test the performance of surfactant injection in the transition zone and mature waterflood areas (Pal, et.al, 2018, Pal, et.al, 2019, Kumar, et.al. 2024, Kumar, et.al.2025). In addition to these studies, the application of wettability-altering surfactants was reported in different carbonate reservoirs, including Semoga field in Indonesia (Rilian et al. 2008), tight U.S. shale reservoirs (Ataceri, 2023), and the Baltic Basin reservoirs (Dangi et al., 2024), all showing promising results.

Given the encouraging outcomes observed from the application of wettability-altering surfactants in both laboratory experiments and field trials, a comprehensive two-phase study was initiated to explore the potential of surfactant-based chemical enhanced oil recovery (EOR) techniques in high-salinity, high-temperature carbonate reservoirs managed by PDO. This strategic investigation aimed to optimize the selection and deployment of surfactants tailored to the challenging reservoir conditions. The two-phase methodology was designed to systematically identify suitable surfactants: Phase 1 focused on screening and evaluating a diverse range of surfactants for their compatibility with the reservoir's brine, ensuring they could withstand the harsh salinity and temperature environments. Following this, Phase 2 involved more advanced testing, including emulsion stability assessments and core flood experiments, to assess the surfactants' effectiveness in altering wettability and improving oil displacement under reservoir-like conditions. This extensive experimental program, spanning over a year, involved the rigorous screening of more than 10 surfactant candidates. Through meticulous analysis and iterative testing, the study culminated

in the identification of four surfactants deemed highly promising for field-scale application, offering a robust foundation for future EOR implementation in PDO's carbonate reservoirs.

Surfactant Screening

For EOR screening two carbonate reservoirs were considered with their properties shown in Table 1. Reservoir 1 is Natih carbonate and Reservoir 2 is Lower Shuaiba carbonate. Both fields show low-moderate reservoir rock permeability (2-20 mD) and strong oil-wet tendencies—ideal for wettability-altering chemicals. Although Reservoir 2 exhibits more challenging conditions, the task is to determine a surfactant formulation that performs effectively across both reservoirs. The surfactant selection process considered various surfactants' stability under field conditions, with specific focus on temperature and salinity tolerances. Salinity range of 120-180K ppm and temperature range of 60-85 °C were considered. Several wettability altering surfactants, which have applicability in the given temperature and salinity range were investigated, see Figure 1.

Table 1—Table summarizing Reservoir properties.

Reservoir conditions	Reservoir 1 (Natih)	Reservoir 2 (Lower Shuaiba)
Temperature, °C	60	85
Brine salinity, ppm	120,000	180,000
Oil viscosity, cP	2.5	5-6
Permeability, mD	2-20	

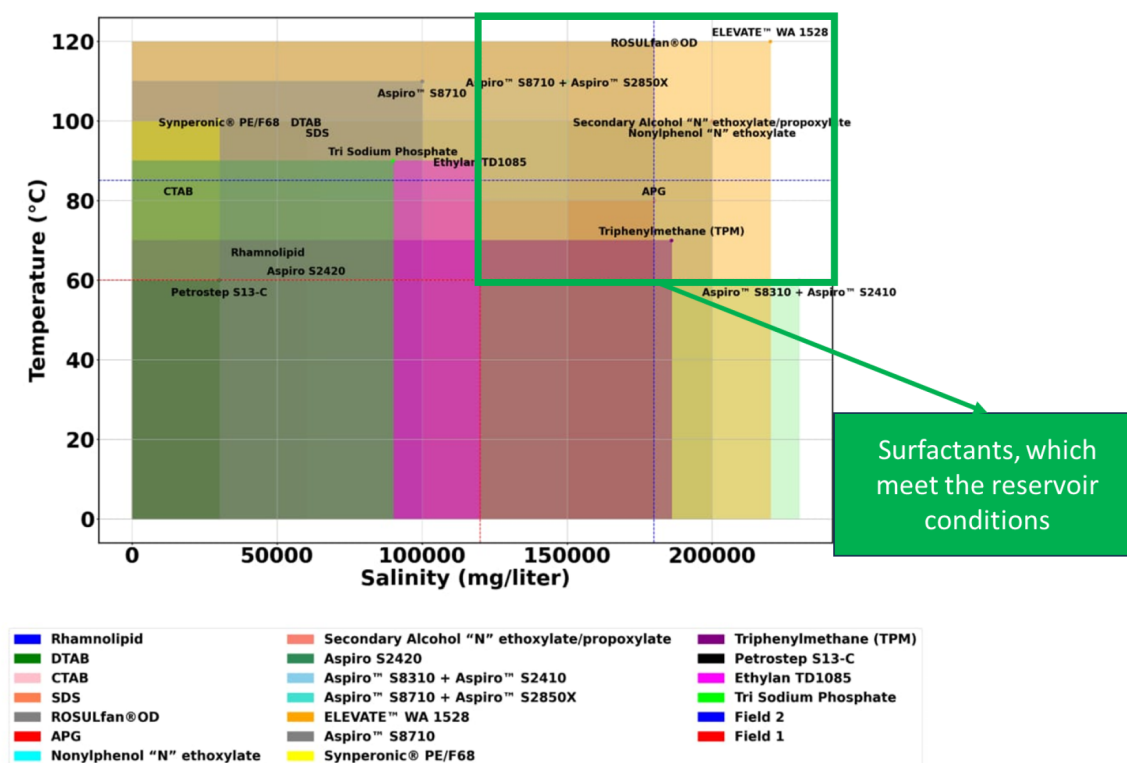


Figure 1—Figure showing Temperature and Salinity based screening of surfactants. Surfactant active in temperature range of 60-85 °C and salinity range of 120-180 K ppm were investigated.

Initial screening based on temperature and salinity range, see [Figure 1](#), identified surfactants like TERGITOL™ 15-S-15, and ELEVATE™ WA 1528, ASPIRO class of surfactants as promising candidates for chemical EOR across both fields. These surfactants exhibited compatibility with the reservoir conditions of Reservoir 1 and Reservoir 2, based on stability tests in high-temperature and high-salinity environments. A total of 10 surfactants were identified and selected for further testing. These included TRINTON-CG 110, TPM, Tergitol-15-S-20, TSP, ASPIRO S2420, NanoSurf, Tergitol-15-S-15, ASPIRO S8310, ASPIRO S6420, and out of these 10 surfactants 4 are recommended for further testing and analysis, which include TRINTON-CG, Tergitol-15-S-20, Tergitol 15-S-20, ASPRIO S2420, Tergitol 15-S-15, ASPIRO S6420. Surfactants are also listed with their applicability ranges, chemical formula, and surfactant manufacturer in [Table 2](#).

Table 2—Table showing the 10 identified surfactant their application range, chemical formula, and surfactant provider.

S. No	Surfactant	Type	Temp (°C)	Salinity, PPM	Selected Surfactants	Field tested	Chemical Formula	Company
1	Aspiro™ S 8710 + Aspiro™ S 2850 X	Nonionic	<110	30000-150000	Yes		N/A	BASF
2	Nonylphenol "N" ethoxylate	Nonionic	100	200,000	Yes		C ₁₅ H ₂₄ O	MERCK
3	Alkyl Polyglycoside (APG) (Triton™ CG-110)	Nonionic	80	180,000	Yes		C ₁₀ H ₂₁ O-(C ₆ H ₁₀ O ₅) _{1.5} H	DOW Chemicals
4	Secondary Alcohol "N" ethoxylate/propoxylate (TERGITOL™ 15-S-15)	Nonionic	100	200,000	Yes	Test for Trials in Qatar by Maersk Oil (Bennetzen, et.al., 2014)	C ₁₂₋₁₄ H ₂₅₋₃₁ O(CH ₂ CH ₂ O) ₁₅ H	DOW Chemicals
5	Triphenylmethane (TPM)	Nonionic	70	186000	Yes		(C ₆ H ₅) ₃ CH	MERCK
6	Aspiro™ S 8310 + Aspiro™ S 2410	Nonionic	<60	6000-23000	Yes		N/A	BASF
7	ELEVATE™ WA 1528 (Tergitol-15-S-20)	Nonionic	120	220,000	Yes	Al-Shaheen Field Trail, Qatar (Pal, et.al, 2018)	C ₁₂₋₁₅ H ₂₅₋₃₁ O(CH ₂ CH ₂ O) ₂₀ H	DOW Chemicals
8	Trisodium Phosphate	N/A	N/A	N/A	Yes	Central Texas Carbonate 1990 (Olsen, et.al., 1990)	Na ₃ PO ₄	MERCK
9	Aspiro S2420	Nonionic	<100	120,000	Yes		N/A	BASF
10	Nano Surf EOR	Nonionic	<100	120,000	Yes		Alcohol Alkoxyate	DOW Chemicals

A series of temperature stability tests was performed to identify surfactants for the reservoir conditions. These experiments involve solubility testing of surfactant at room temperature and at the two different reservoir temperature conditions (60 and 85 °C), as shown in [Figure 2](#). Also, reservoir brine of 120 and 180K ppm salinity was tested. Two different surfactant concentrations were tested: 2,500 and 5,000 ppm. The choice of surfactant concentration is based on the successful application of surfactant in a similar range in field trials reported in the US and Middle Eastern carbonates ([Yang and Wadleigh, 2000](#), [Valdez et.al., 2020](#), [Bennetzen, et.al, 2014](#), ([Pal, et.al, 2018](#), [Pal, et.al, 2019](#), [Kumar, et.al. 2024](#), and [Kumar, et.al.2025](#)). Most field applications have used surfactant concentration in the range of 2500 to 5000 ppm. The temperature

and reservoir salinity testing showed that out of the total 10 surfactants, 5 showed good stability at HSHT conditions, as summarized in Figure 3. These include Triton-CG 110, Trisodium Phosphate (TSP), Aspiro S2420, 4- N, and Aspiro S6420. Surfactants that work for both reservoirs are Triton-CG 110, Aspiro S2402, and Aspiro S6420. Final selection of surfactant for field application is based on cost and previous use of such chemicals in field trials.

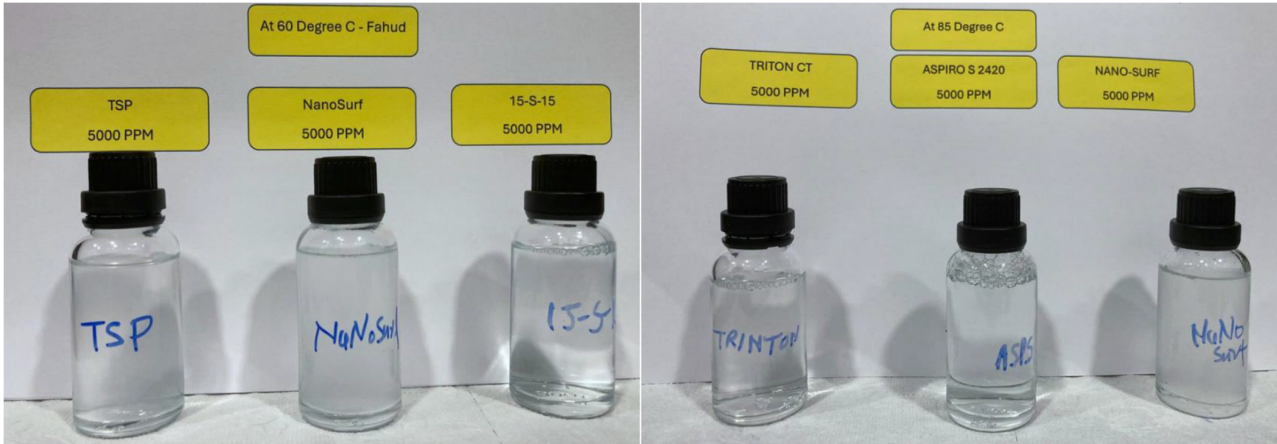


Figure 2—Figure showing temperature stability test at two different temperatures at 60°C and 85° Condition and at field condition at 5000 PPM.

	Surfactants	TRINTON-CG 110	TPM	15-S-20	TSP	ASPIRO S2420	NanoSurf	15-S-15	4-N	ASPIRO S8310	ASPIRO S6420
		5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM
Reservoir 1	Clouding at room temp	No	Yes	No	Yes	No	Yes	No	Slight	Yes	No
	Clouding at 60°C	No	Yes	No	Yes	No	Yes	No	Yes	Yes	No
	Surfactant	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM
	Clouding at room temp	No	Yes	No	Yes	No	Yes	No	No	Yes	No
	Clouding at 60°C	No	Yes	No	Yes	No	Yes	No	No	Yes	No
Reservoir 2	Surfactants	TRINTON-CG110	TPM	15-S-20	TSP	ASPIRO S2420	NanoSurf	15-S-15	4-N	ASPIRO S8310	ASPIRO S6420
	Concentration tested	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM	5000 PPM
	Clouding at room temp	No	No	No	No	No	Yes	No	No	Yes	No
	Clouding at 85 oC	No	Yes	Yes	No	No	Yes	Yes	No	Yes	No
	Concentration tested	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM	2500 PPM
	Clouding at room temp	No	No	No	No	No	Yes	No	No	Yes	No
	Clouding at 85 oC	No	Yes	Yes	No	No	Yes	Yes	No	Yes	No

Figure 3—Figure showing final selection of surfactant, which were stable at high temperature and high salinity conditions, Top part of the table shows results for Reservoir 2 and bottom part of the table shows results for Reservoir 1.

The Phase 1 findings highlight the potential of selected surfactants to address oil recovery challenges from Reservoir 1 and Reservoir 2, enhancing oil production rates and recovery efficiency. With surfactants such as Triton-CG 110, Aspiro S2420, and Aspiro S6420 demonstrating stability under high temperatures and salinity, the project will proceed with fluid-fluid testing to further assess recovery performance in a simulated reservoir environment. Continued testing and coreflooding will refine these selections and support optimized chemical EOR strategies in Oman's carbonate fields, ultimately contributing to sustainable resource utilization and improved reservoir performance.

Emulsion Testing

In this section we present the phase behavior testing, first step of which includes testing for any emulsion formation using 30-70 crude and brine mixture with surfactants. The emulsion testing is important because it helps to identify impact of surfactant on flow behavior and surfactant efficiency. Surfactant injection could lead to the formation of favorable or unfavorable emulsions due to changes in IFT between oil and water. These emulsions, particularly oil-in-water (O/W) emulsions, could improve oil recovery by enhancing sweep efficiency and reducing the mobility of injected water. On the other hand, formation of high viscosity emulsions could negatively impact the flow behavior by blocking pores and needs to be avoided when choosing the surfactant, as it can lead to reduced injection and even injectivity impairment. For details on the impact of surfactant-induced emulsions on flow behavior in reservoirs please see ([Bashir et.al, 2022](#)).

Separation crude oil/brine system: room temperature

The emulsion tendency testing was first carried out at room temperature, which was set around 20° C. Tests were carried out with increasing surfactant concentration to test for the stability of any emulsions present. The first test was conducted for Reservoir 1, followed by testing for Reservoir 2. For this test, 3 surfactants were used, Triton CG-110, Aspiro S2420 and Aspiro S6420. Reservoir 1 oil-brine mixtures were tested with these 3 surfactants at concentrations from 100 to 10,000 ppm, with stability evaluated over 24 hours. In all cases, no microemulsion phase was detected (see [Figure 4](#) for the results with Aspiro S2420). Similarly, Reservoir 2 oil-brine mixtures were tested with Triton CG-10, Aspiro S2420, and Aspiro S6420 at concentrations ranging from 100 to 10,000 ppm, with stability evaluated over 24 hours. In all cases, no microemulsion phase was detected (see [Figure 5](#) for the results with Aspiro S2420). These results indicate that none of the three surfactants formed microemulsions under the tested conditions.

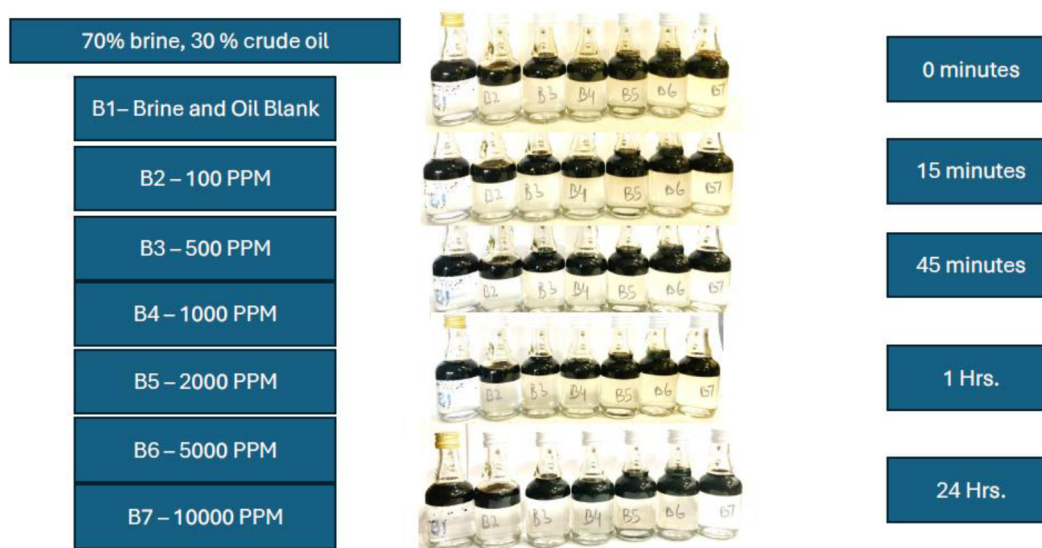


Figure 4—Surfactant Aspiro S2420 used for emulsion stability test with Reservoir 1 oil and Brine.

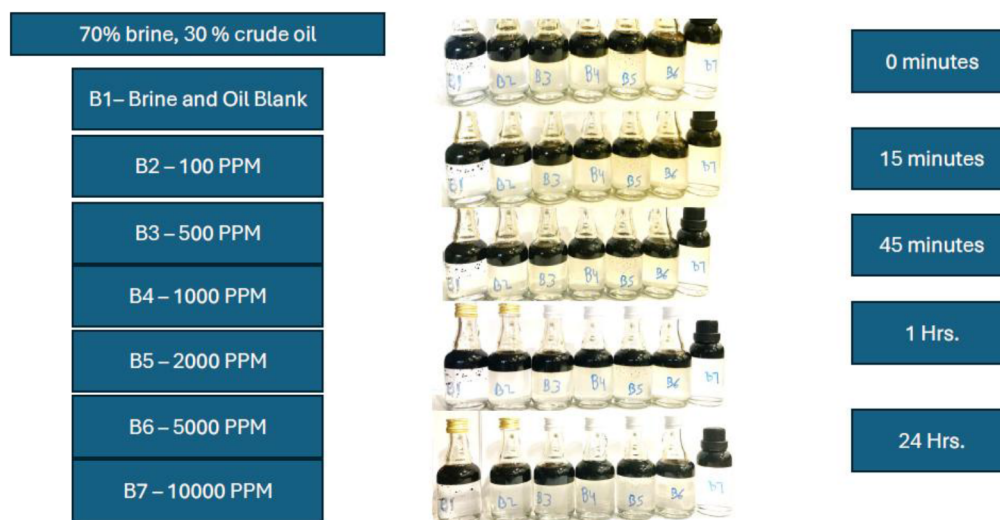


Figure 5—Surfactant Aspiro S2420 used for emulsion stability test with Reservoir 2 oil and Brine.

Separation crude oil/brine system: reservoir temperature

In the context of surfactant screening for enhanced oil recovery (EOR), specific tests such as separation of the crude oil/brine system and microemulsion decay over time at reservoir temperature are conducted to evaluate surfactant performance under realistic conditions. The separation test assesses the surfactant's ability to stabilize or destabilize emulsions, which is critical for determining its efficiency in mobilizing oil and maintaining injectivity, as stable emulsions can lead to operational challenges like clogging (Sheng, 2014). Microemulsion decay testing at reservoir temperature (e.g., 60-85°C for HST reservoirs) measures the longevity and stability of the surfactant-formed microemulsions, which are essential for sustained low interfacial tension and wettability alteration over the injection period. These tests ensure the selected surfactants remain effective under reservoir-specific thermal and compositional stresses, guiding optimal formulation for field deployment. In this phase of testing an additional surfactant was included, i.e., Tergitol-15-S-20, which showed promising results for Reservoir 1 only. This surfactant has already been used in some field trials giving excellent results.

Test procedure phases:

- Conditioning the crude oil at the temperature of the planned measurement.
- Introduction of 40 ml of crude oil and 40 ml of water/brine into a 100 ml measuring cylinder.
- Conditioning the cylinder for 30 minutes in a bath at the set temperature.
- Homogenizing the contents of the cylinder with a mechanical stirrer for 5 minutes at 1500 rpm.
- Dosing of the surfactant into the produced emulsion so that its concentration reaches the set level.
- Homogenization of the emulsion - stirring at 100 rpm for 1 minute.
- Evaluation of the amount (ml) of the emitted phases at the end of the mixing of oil, emulsion, and water at the preset stand-off intervals.

At this stage of the study, dead crude and brine with a 50%/50% ratio for both reservoirs were taken. Brine contained four different surfactants - Aspiro S2420, Aspiro S 6420, Tergitol-15-s-20, Triton CG-110 with a 5000-ppm concentration. The evaluation was carried out against the dead crude/brine system (same ratio 50%/50%) containing no surfactant. During the phase separation analysis, it was evaluated and summed with the oil phase. From the results shown in the figures below it seems that all surfactants tested at 60°C (Reservoir 1) and at 85°C (Reservoir 2) allowed the oil/water dispersion system to accelerate separation.

The separation curves are shown in Figure 6 (Reservoir 1) and Figure 7 (Reservoir 2). The most effective surfactants were Triton CG-110 and Aspiro S2420; separation of more than 90% of the brine was achieved after just 30 minutes. It should be noted that the aqueous phase contained dispersed oil in all cases; the aqueous phase was cloudy, while it clarified most readily in the case of surfactant Triton CG-110 followed by Aspiro S2420. After 24 hours, effective complete separation (almost complete clarification of the aqueous phase) was observed for surfactants Aspiro S2420 and Triton CG-110, which were selected for further testing.

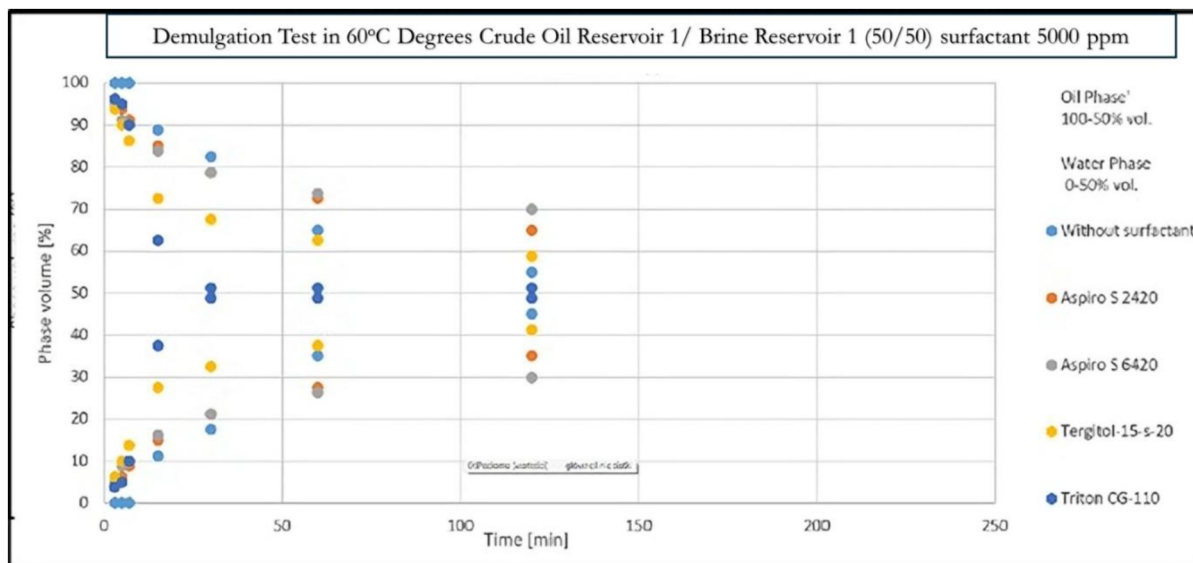


Figure 6—Demulcation test for Reservoir 1 at concentration 5000 ppm and 60°C

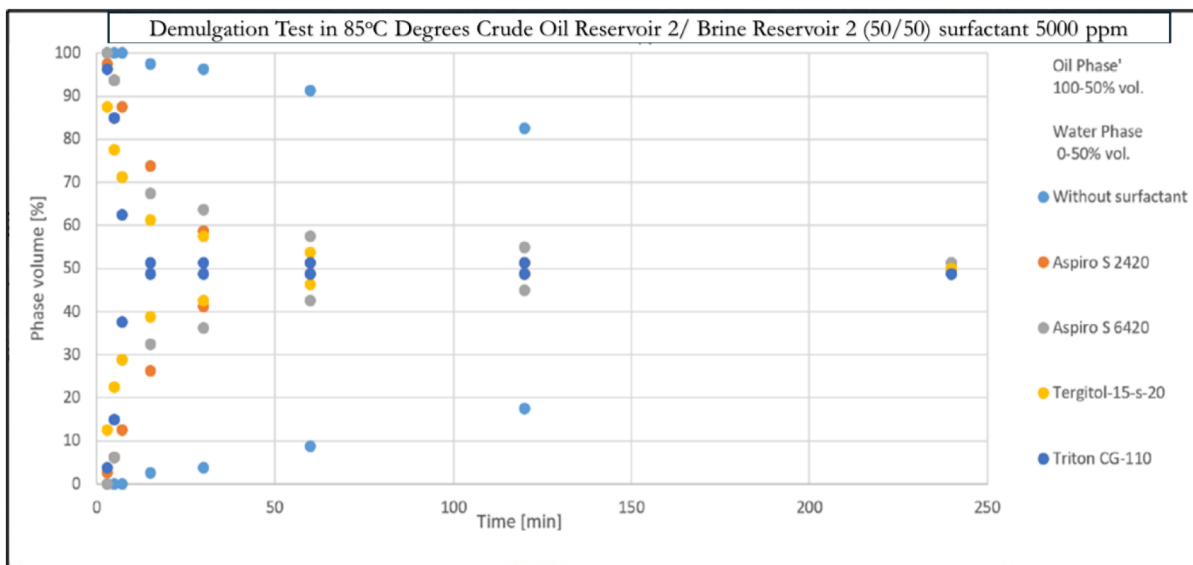


Figure 7—Demulcation test for Reservoir 2 at concentration 5000 ppm and 85°C

In a further series of tests, surfactants Aspiro S2420 and Triton CG-110 dosed at 2500 and 1000 ppm were evaluated under conditions analogous to the initial tests. The separation curves are shown in Figure 8 (Reservoir 1) and Figure 9 (Reservoir 2). In the first stage of separation (<10 min), the separated aqueous phase was cloudy due to the hydrocarbons suspended in it, while after 60 min, a clarification process of the aqueous phase was observed. For Triton CG-110, it was found that the separation quality depended on its

concentration and improved with increasing concentration. In all cases, the separation efficiency of more than 90% was obtained after only 30 minutes. However, Triton CG-110 shows a tendency to foam. In the case of surfactant Aspiro S2420, a similar relationship between concentration and separation efficiency was observed. However, with the lowest dosage (1000 ppm), after 30 minutes the separation efficiency reaches ~80%. Still the observed separation was faster than in the system containing no surfactant. It should be noted that for the other two surfactants i.e., Aspiro S6420 and Tergitol 15-s-20, the aqueous phase was brown in color and characterized by strong turbidity indicating a high content of dispersed hydrocarbons. Based on these results, it was confirmed that both surfactants Triton CG-110 and Aspiro S2420 demonstrate good separation efficiency even in the wide range of surfactant concentration, Therefore, these surfactants were selected for further testing.

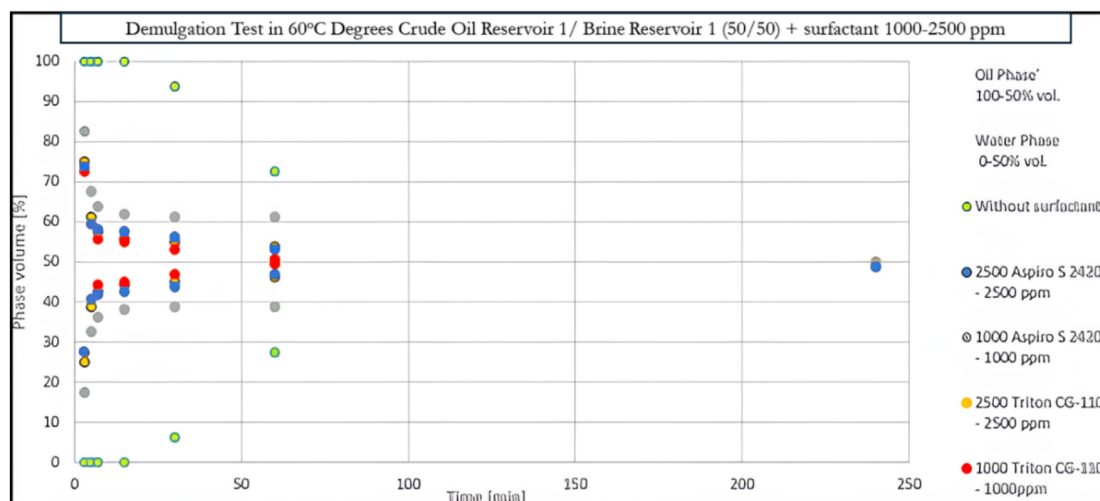


Figure 8—Demulgation test for Reservoir 1 at concentration 2500 ppm and 1000 ppm.

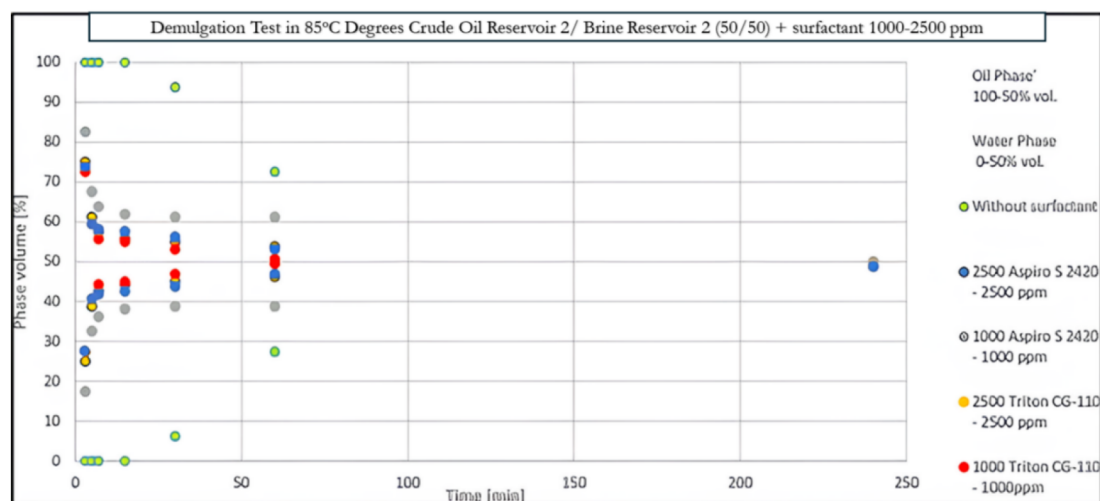


Figure 9—Demulgation test for Reservoir 2 at concentration 2500 and 1000 ppm

Measurements of Contact Angle and Interfacial Tension

Contact angle measurement is one of the most illustrative methods for characterizing rock wettability. In the present experiments, field core material, crude oil, and reservoir brine were used. Polished rock surfaces were prepared from core cuttings using a 600-mesh (or finer) grinding plate, then aged with formation fluids at reservoir temperature. The samples were subsequently placed in a visualization cell filled with brine.

Oil droplets were deposited at random locations on the rock surface, and their shapes were analyzed to determine contact angles using an optical contact angle measuring and contour analysis system (OCA 25-PMC, DataPhysics). Each test was repeated several times under reservoir conditions, and averaged values are reported in Table 3. Oil-water interfacial tension (IFT) was measured by the pendant drop method using the same OCA 25-PMC system. Reported IFT values represent averages from at least three measurements (Table 3).

Table 3—Contact angle and IFT measurements for the cases with and without surfactant.

Measurements		Reservoir 1 (Natih)	Reservoir 2 (Lower Shuaiba)
Contact angle, °*	w/o surfactant	60	80
	w/ surfactant Aspiro S2420	43	45
Interfacial tension, mN/m*	w/o surfactant	14.4	28.4
	w/ surfactant Aspiro S2420	1.9	1.5

* Values as an average of at least 3 measurements

Both sets of measurements confirm that the presence of surfactant (Aspiro S2420) alters rock wettability under Reservoir 1 and Reservoir 2 conditions, as evidenced by changes in both contact angle and oil-water interfacial tension.

Core Flooding Experiments

The core flooding experiments shown here were carried out using field core plugs obtained from the wells cored in Reservoir 1 and Reservoir 2. In this phase of testing, based on the results from the previous phase, it was decided to use SPIRO S2420 as the surfactant for core flooding tests. The core flooding experiments were designed to simulate two distinct reservoir conditions:

- **Scenario 1** - Far-field conditions which represents reservoir regions away from the wellbore, where the reservoir has not been exposed to large amount of injected water and is unlikely to have reached residual oil saturation. To replicate these conditions, the core flooding procedure involves injecting of only 0.5 pore volumes (PV) of water before introducing the surfactant.
- **Scenario 2** - Near-wellbore conditions which represents the reservoir zone close to the wellbore, where the rock has been exposed to a large volume of injected water and is more likely to have reached residual oil saturation. In this case, the core flooding procedure involves injecting 2.5-3.5 PV of water before the surfactant is introduced. Additionally, after the surfactant injection a soaking period of 24hrs is considered before chase water is injected. The reason for soaking period is to mimic kind of surfactant induced huff and puff type injection test in a producer well.

Core ageing with reservoir dead crude oil was carried out at reservoir temperature (60°C for Reservoir 1 and 80°C for Reservoir 2) using an autoclave placed in a heating chamber. The ageing process lasted for over 60 days. After the aging, the core flooding tests were carried out in three steps:

Brine injection (step 1)

- Pumping brine through the sample with a flow rate of 0.114 cm³/min - this should correspond to a slow velocity of brine flow in reservoir.
- Measuring oil saturation after stabilization of the brine flow, and no more oil is observed in the output downstream of the sample.
- Measuring effective permeability for brine.

Brine and surfactant mixture injection and soaking (step 2)

- Pumping a mixture of brine and surfactant (Aspiro S2420) through the sample at a flow rate of 0.114 cm³/min. The concentration of surfactant was 10,000 ppm. - but the concentration of the active component was 5,000 ppm.
- In case core flood experiment was executed as per scenario 2 the core saturated with surfactant was soaked for 24 hours.

Chasing Brine injection (step 3)

- Pumping brine through the sample with a flow rate of 0.114 cm³/min.
- Measuring oil saturation after stabilization of the brine flow, and no more oil is observed in the output downstream of the sample.
- Measuring effective permeability for brine.

Results of Core flooding Experiments: Reservoir 1

A total of 3 coreflooding test were conducted for Reservoir 1. The first core flooding test was conducted mimicking the scenario 1. In the first core flood test, initially 0.5 PV of water was injected, followed by 0.5 PV of diluted surfactant, which was followed by 3 PV of chase water. The core flood resulted in an incremental recovery factor of 14% and an ultimate recovery factor of 70% (see Figure 10). The second core flooding test was conducted for Reservoir 1 rock samples mimicking the scenario 2. Initially, 2.5 PV of water flood was carried out on the rock sample, followed by 1 PV of diluted surfactant, and then 3 PV of chase water. The core flood resulted in an incremental recovery factor of 14% and an ultimate recovery factor of 78% (see Figure 11).

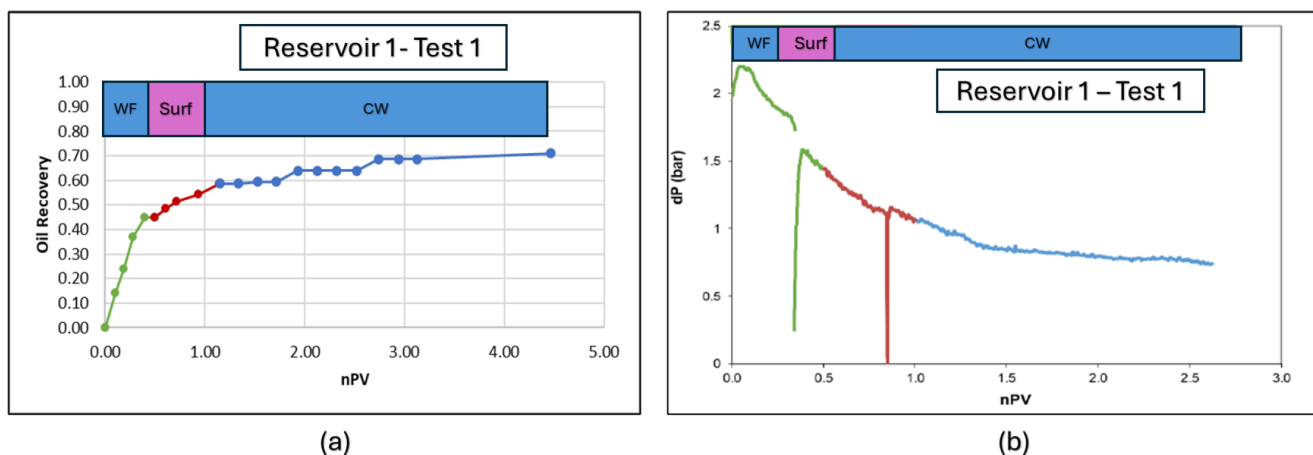


Figure 10—Reservoir 1 Test 1- a) Figure showing the core flood conducted for Reservoir 1 with 0.5 PV of WF followed by 0.5 PV of diluted surfactant and 3PV of chase water. b) Differential pressure changes during initial waterflood, surfactant injection and chase water (CW) injection.

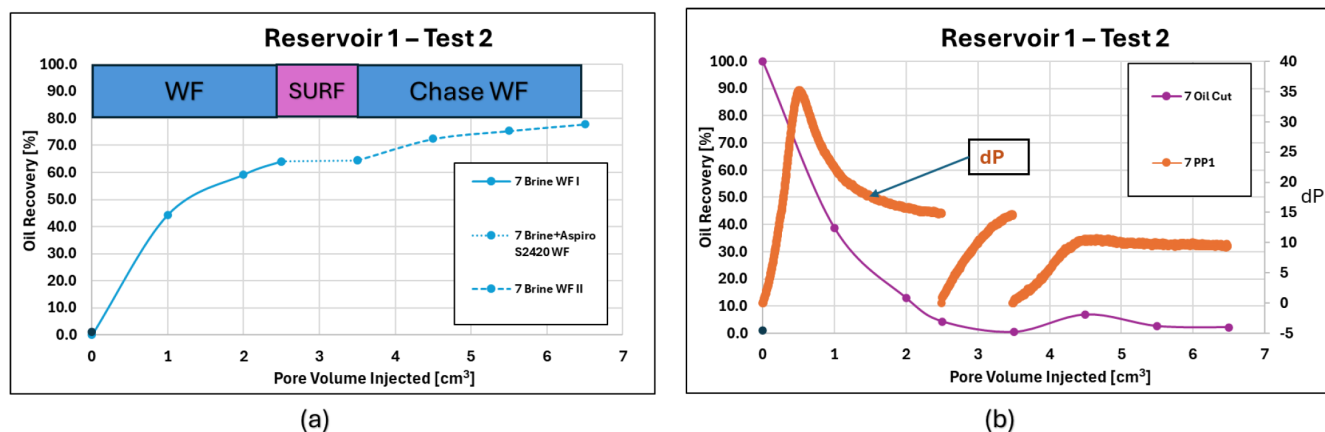


Figure 11—Reservoir 1 Test 2- a) Figure showing the core flood conducted for Reservoir 1 with 2.5 PV of WF followed by 1 PV of diluted surfactant and 3PV of chase water. b) Differential pressure changes during initial waterflood, surfactant injection and chase water (CW) injection

The 3rd core flooding test for Reservoir 1 was like the second core flooding test, except that this time, 3.5 PV of water flood was carried out on the rock sample, followed by 1 PV of diluted surfactant, which was followed by 3 PV of chase water. The core flood resulted in an incremental recovery factor of 23% and an ultimate recovery factor of 74% (see Figure 12).

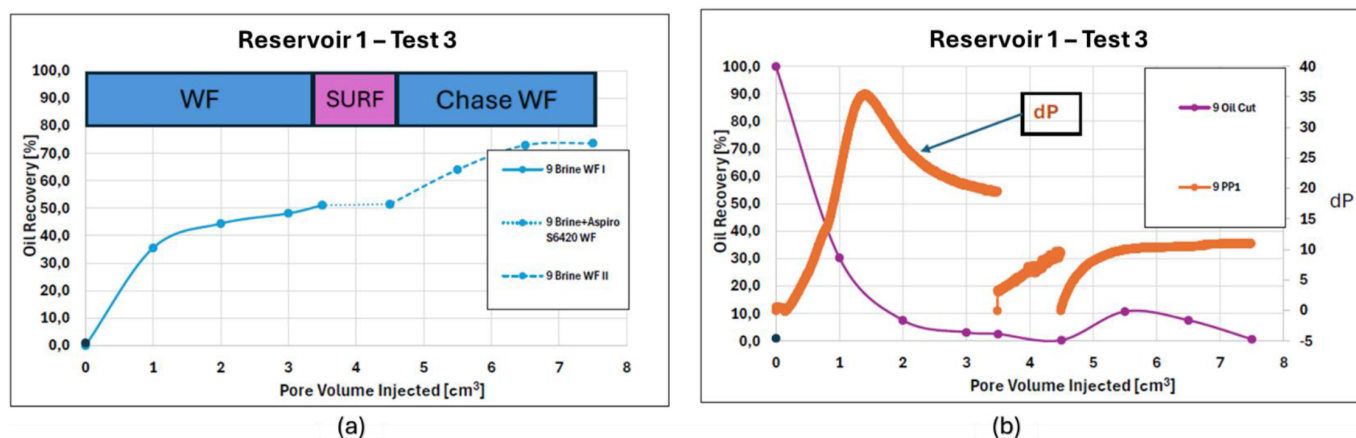


Figure 12—Reservoir 1 Test 3 a) Figure showing the core flood conducted for Reservoir 1 with 3.5 PV of WF followed by 1 PV of diluted surfactant and 3PV of chase water. b) Differential pressure changes during initial waterflood, surfactant injection and chase brine injection

Results of Core flooding Experiments: Reservoir 2

A total of 2 coreflooding tests were conducted for Reservoir 2. Both tests mimicked core flooding scenario 2, which represents nearwell-bore reservoir. In the first test, initially 3 PV of water flood was carried out on the rock sample, followed by 1 PV of diluted surfactant (soaked for 24 hours), which was followed by 3PV of chase water. The core flood resulted in an incremental recovery factor of 11% and an ultimate recovery factor of 70% (see Figure 13).

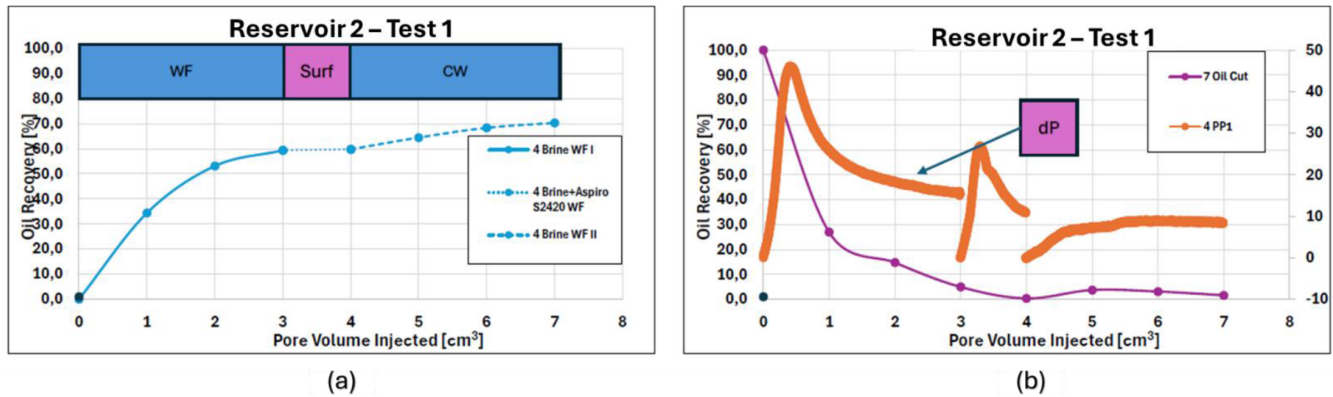


Figure 13—Reservoir 2 Test 1 a) Figure showing the core flood conducted for Reservoir 1 with 3.5 PV of WF followed by 1 PV of diluted surfactant and 3PV of chase water. b) Differential pressure changes during initial waterflood, surfactant injection and chase brine injection

In the second test, initially 3.5 PV of water flood was carried out on the rock sample, followed by 1 PV of diluted surfactant, which was followed by 3PV of chase water. The core flood resulted in an incremental recovery factor of 15% and an ultimate recovery factor of 83%. See the results shown in Figure 14.

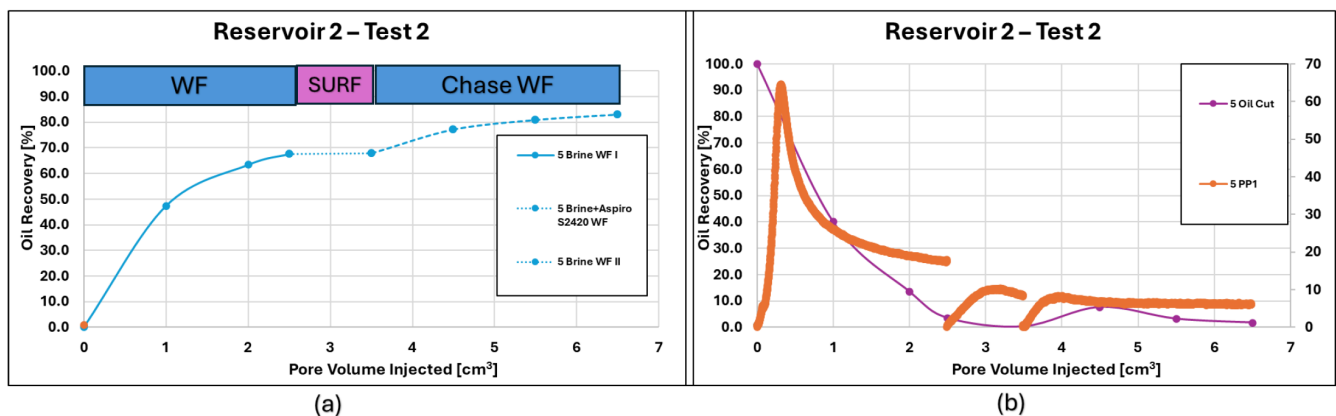


Figure 14—Reservoir 2 Test 2 a) Figure showing the core flood conducted for Reservoir 1 with 2.5 PV of WF followed by 1 PV of diluted surfactant and 3PV of chase water. b) Differential pressure changes during initial waterflood, surfactant injection and chase brine injection

A series of five surfactant core floods was conducted using field-aged carbonate cores. All experiments demonstrated significant incremental oil recovery due to the surfactant injection, with ultimate recovery exceeding 70% (see Table 4). Such ultimate recovery would not be attainable under waterflooding alone due to the strong oil-wet nature of the rock. The enhanced recovery confirms the effectiveness of surfactant flooding in altering wettability and mobilizing residual oil. Variations in the incremental recovery observed among the tests are attributed primarily to differences in field core properties such as permeability, porosity etc. These results highlight the potential of surfactant flooding to substantially improve oil recovery in oil-wet carbonate reservoirs.

Table 4—Results of core flood experiments.

Reservoir	Test No.	Brine permeability, mD	Porosity, %	PV water injected	RF due to water (step 1), %	RF due to surfactant (step 2), %	RF due to chase water (step 3), %	Ultimate RF, %
Reservoir 1 (Natih)	1	7.70	28.6	0.5	46	14	10	70
	2	3.97	26.7	25	64	14		78
	3	3.17	36.4	3	51	23		74
Reservoir 2 (Lower Shuaiba)	1	8.36	37.6	3	59	11		70
	2	7.05	39.1	3.5	68	15		82

Conclusions and Recommendations

- Surfactant screening was successfully conducted for two carbonate reservoirs, resulting in the identification of four suitable candidates. Among these, Aspiro S2420 and Triton CG-10 (non-ionic surfactants) demonstrated the most encouraging performance.
- Aspiro S2420, selected for further evaluation due to its established application in other carbonate EOR projects, showed strong performance by enhancing oil mobility in both reservoirs. Coreflood experiments confirmed significant incremental oil recovery, with ultimate recovery factors exceeding 70% in all cases—substantially higher than those achieved with waterflooding alone—highlighting the effectiveness of the tested surfactants in mobilizing residual oil.
- Variability in incremental recovery among the coreflood experiments was attributed to differences in reservoir properties, particularly permeability, emphasizing the need to tailor surfactant applications to specific reservoir conditions.
- Overall, the results demonstrate the strong potential of non-ionic surfactants as effective chemical EOR agents capable of extending the productive life of mature reservoirs. Recovery improvements of more than 10% under challenging conditions could yield substantial additional oil volumes and economic benefits.
- While Aspiro S2420 is recommended for field application and prioritized due to its proven track record, Triton CG-10 also exhibited promising performance during screening and merits further evaluation in future coreflood testing phases.

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